RAJALAKSHMI ENGINEERING COLLEGE

Alagappan R 251801018-2025-AIDS

A2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 10:41 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:41 AM |
| **Time taken** | 39 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    long long int_a = *((long long *)a);
    long long int_b = *((long long *)b);
    if (int_a < int_b) return -1;
    if (int_a > int_b) return 1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    long long *arr = (long long *)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
    }

    // Sort the array in ascending order using greedy strategy
    qsort(arr, n, sizeof(long long), compare);

    long long max_sum = 0;
    for (int i = 0; i < n; i++) {
        max_sum += arr[i] * i;
    }

    printf("%lld\n", max_sum);

    free(arr);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>5<br>3<br>4<br>0 | 40 | 40 | ✔ |
| ✔ | 10<br>2<br>2<br>2<br>4<br>4<br>3<br>3<br>5<br>5<br>5 | 191 | 191 | ✔ |
| ✔ | 2<br>45<br>3 | 45 | 45 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course